Math 526, S16 **Name:** Answer

Quiz 1 **Section:** **SSN:** xxx-xx-

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (7 points) Given $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (1, 5)$.

(a) Find the *cosine* of the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\|V\| = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\|W\| = \sqrt{1^2+5^2} = \sqrt{26}$$

$$V \cdot W = 1\cdot 1 + 1 \cdot 5 = 1+5 = 6$$

$$\Rightarrow \cos\theta = \frac{V\cdot W}{\|V\|\cdot\|W\|} = \frac{6}{\sqrt{2}\sqrt{26}} = \frac{6}{\sqrt{52}} = \frac{6}{2\sqrt{13}} = \frac{3}{\sqrt{13}}$$

(b) Choose a number c so that $\mathbf{w} + c\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$W + CV$ is perpendicular to V

$$\Rightarrow (W + CV) \cdot V = 0$$

$$W \cdot V + (cV) \cdot V = 0$$

$$W \cdot V + c(V \cdot V) = 0$$

$$C = -\frac{W \cdot V}{V \cdot V}$$

$$= -\frac{(1,5)\cdot(1,1)}{(1,1)\cdot(1,1)}$$

$$= -\frac{6}{2}$$

$$= -3$$

Problem 2. (3 points) Compute $\mathbf{Ax}$ (i.e., the matrix $\mathbf{A}$ acting on the vector $\mathbf{x}$) with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 0 & -1 \\ -4 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

by using linear combination of the columns of $\mathbf{A}$ with components of $\mathbf{x}$ as coefficients.

$$Ax = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 0 & -1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -4 \\ -8 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2+4+0 \\ -4+0-3 \\ -8+2+6 \end{bmatrix} = \begin{bmatrix} 6 \\ -7 \\ 0 \end{bmatrix}$$